MAJD KHALAF

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EDUCATION

Northeastern University Boston, MA
Ph.D. in Computer Engineering, Dept. of Electrical & Computer Engineering September 2025 – May 2030 (expected)

- GPA: 4.00/4.00 | Advisor: Prof. Mallesham Dasari
- Research interests: Wireless sensing and communication for AR/VR and smart glasses; low-power wearable sensing; signal processing and machine learning for sensor data; hardware prototyping and hardware/software co-design

Norwich University Northfield, VT
B.S. in Electrical & Computer Engineering *August 2021 – May 2025*

- GPA: 3.97/4.00 | Minors: Mathematics and Computer Science

RESEARCH EXPERIENCE

Northeastern University, SINRG Lab Boston, MA
Graduate Research Assistant *September 2025 – Present*

- Research wireless and wearable sensing for AR/VR smart glasses, from sensor and hardware prototyping through signal processing, on-device inference, and wireless connectivity
- **CapGaze (ACM SenSys 2027):** Designed a camera-free sensing modality for eye tracking using periocular capacitive electrodes and a reflective photosensor; built the hardware prototype and runs real-time gaze estimation on an embedded MCU
- Developed the end-to-end signal-processing and gaze-estimation pipeline, including a two-pass self-correction algorithm; designed and ran an IRB-approved data collection study with 15 participants
- **Smart-glasses measurement study (ACM MobiCom 2026):** Led the first characterization of multimodal applications on commercial XR/AI glasses, measuring end-to-end latency, throughput, and rail-level power over Bluetooth/BLE, Wi-Fi Direct, and 5G; published and presented the work at MobiCom 2026
- **Drakon (ACM MobiCom 2026 WiP):** Developing an open smart-glasses prototype with native LTE-M cellular connectivity
- Collaborate across lab projects on volumetric video streaming (real-world network-trace evaluation) and immersive VR training (user studies)

Norwich University AI Center Northfield, VT
AI Research Fellow & Teaching Assistant *January 2024 – May 2025*

- Developed machine learning and NLP models (e.g., a genetic-algorithm-optimized Bi-LSTM) for misinformation detection, digital forensics, and text analysis, contributing to five peer-reviewed publications
- Delivered lectures, tutorials, and lab content for a senior-level AI course with 20+ students
- Mentored four French military college students on AI-based capstone projects in a cross-cultural, international collaboration

PUBLICATIONS & MANUSCRIPTS

Accepted

- **M. Khalaf**, G. Chen, A. Hanson, M. Dasari. "CapGaze: Capacitive Sensing-based Eye Tracking on XR and AI Smart Glasses," *ACM SenSys*, 2027 (accepted).

Under Review

- **M. Khalaf**, T. Olukanni, A. Al Bataineh. "AI-Powered Smart Glasses for Visually Impaired Users," submitted to *IEEE Transactions on Human-Machine Systems*.
- B. Balamurugan, S. In, **M. Khalaf**, R. Dahiya, D. Shakthivel, E. McGivney, M. Dasari. "Immersive Workforce Training for Tool-Mediated, Pressure-Sensitive Object Manipulation: A Silicon Wafer Study," submitted to *IEEE VR*, 2027.

Published

- **M. Khalaf**, M. Garcia, M. Chadha, M. Dasari. "A First Look at Multimodal Mobile Application Performance on XR and AI Glasses," *ACM MobiCom*, 2026.
- **M. Khalaf**, M. Dasari. "Drakon: An Open Smart-Glasses Prototype with Native LTE-M Connectivity," *ACM MobiCom (Work-in-Progress)*, 2026.
- **M. Khalaf**, T. C. Olukanni, D. M. Feinauer, M. Cross, A. Al Bataineh. "Future-Ready Students: Validating the Use of Natural Language Processing to Analyze Student Reflections on a Remote Learning Group Project," *ASEE Annual Conference & Exposition*, 2025.
- J. Adkins, A. Al Bataineh, **M. Khalaf**. "Identifying Persons of Interest in Digital Forensics Using NLP-Based AI," *Future Internet*, 16(11):426, 2024. doi:10.3390/fi16110426
- T. C. Olukanni, **M. Khalaf**, M. Cross, D. M. Feinauer, A. Al Bataineh. "Future-Ready Students: Survey Analysis Utilizing Natural Language Processing," *FYEE*, Boston, MA, 2024. doi:10.18260/1-2–48593
- **M. Khalaf**, D. M. Feinauer, M. Cross. "Future-Ready Students: Providing Opportunities for Remote Collaboration on an Engineering Design Project," *ASEE/IEEE FYEE*, 2024.
- A. Al Bataineh, V. Reyes, T. Olukanni, **M. Khalaf**, A. Vibho, R. Pedyuk. "Advanced Misinformation Detection: A Bi-LSTM Model Optimized by Genetic Algorithms," *Electronics*, 12(15):3250, 2023. doi:10.3390/electronics12153250

INDUSTRY EXPERIENCE

Atombeam Technologies — Moraga, CA
Software Engineer II — *February 2025 – September 2025*

- Built and maintained cloud infrastructure using Infrastructure-as-Code and optimized data pipelines for compression algorithms
- Contributed to observability and monitoring for distributed systems and improved CI/CD automation

Walmart Advanced Systems & Robotics — Andover, MA
Site Reliability Engineer Intern — *June 2024 – August 2024*

- Automated deployments on Linux VMs with Ansible, Docker, and Azure CLI
- Containerized and deployed Datadog Private Location agents for synthetic testing and visibility
- Optimized a Golang documentation viewer using goroutines and semaphores for scalability

TECHNICAL SKILLS

Wireless & RF: Wi-Fi, Bluetooth/BLE, 5G, LTE-M, RF/antenna systems, OTA testing, near/far-field transforms, TCP/IP, QoS, Wireshark

Signal Processing & Sensing: Capacitive sensing, wearable sensing, eye tracking, signal conditioning and filtering, feature extraction, multi-sensor fusion, Kalman and particle filtering, energy profiling

Hardware & Prototyping: Electrode and analog front-end design, ESP32, Teensy, MSP430, Arduino, Raspberry Pi, SPI/I2C/UART/GPIO, LTSpice, Intel Quartus, oscilloscopes, signal generators

Machine Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, NLTK, NumPy, Pandas; deep learning, NLP, real-time on-device inference

Programming: MATLAB, Python, C/C++, Golang, Bash/Shell, Verilog/VHDL

Cloud & DevOps: AWS, Azure, Terraform, Docker, Ansible, CI/CD, Linux, Git

HONORS & MEMBERSHIPS

- Norwich University Student Engineer of the Year, February 2025
- Professor Carol E. Stephens Memorial Award, April 2024
- President, Eta Kappa Nu (Electrical Engineers Honor Society), April 2023 – May 2025
- Vice President, Tau Beta Pi (Engineering Honors Society), October 2023 – May 2025